Q5. (7 points) OPTIMIZATION

Consider the three following tables for an airport database and all attributes are neither indexed nor sorted.

- AIRPLANES (aid, brand, size), aid is the primary key.
- PILOTS (pid, name, age), pid is the primary key.
- LastFlight (aid, pid, date), aid and pid are a composite primary key.

And we want to execute the following SQL query:

```
SELECT P.name
FROM AIRPLANES A, PILOTS P, LastFlight L
WHERE A.aid = L.aid AND P.pid = L.pid
AND P.age < 35 AND A.brand = 'Boeing 737';
```

Assuming:

- There are 1,000 rows in AIRPLANES, 1,000 rows in PILOTS and 1,000,000 rows in LastFlight.
- PILOTS.age ranges from [30 to 49] (both inclusive) equally distributed in PILOTS.
- AIRPLANES.brand has 100 distinct values equally distributed in AIRPLANES.
- LastFlight has every combination of aid and pid.

Suppose the cost of running a SELECT operation is the number of rows in the source table and the cost of running a JOIN operation (Cartesian product) is the total rows of the two source tables. If we execute the query with following access plan, the cost will be 1,001,001,002,000.

| Step | Operation | Cost | Estimated result rows |
|---|---|---|---|
| A1 | Cartesian product (A, L) | 1,001,000 | 1,000,000,000 |
| A2 | Cartesian product (A1, P) | 1,000,001,000 | 1,000,000,000,000 |
| A3 | Select rows in A2 with all conditions | 1,000,000,000,000 | 2,500* |

* Here is how the number of resulting rows were estimated:
- The possibility of A.aid = L.aid is 1/1,000 for there are 1,000 different aid.
- The possibility of P.pid = L.pid is 1/1000 for there are 1,000 different pid.
- The possibility of an airplane brand = 'Boeing 737' is 1/100 for there are 100 different brands.
- The possibility of P.age < 35 is 5/20.
- Since all conditions are independent, the number of resulting rows in A3 is about:

$$1{,}000{,}000{,}000{,}000 * (1/1{,}000) * (1/1{,}000) * (1/100) * (5/20) = 2{,}500.$$

Do you have a better access plan to execute the query with a lower total cost?
Please fill the following form about your access plan.

- You don't have to fill all rows depending on how many steps are in your access plan.
- There should be enough room in each cell for you to answer and make corrections.